

GROUP WORK PROJECT MScFE 600: FINANCIAL DATA

Group Number:11269

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an “X” above).

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Report

Task 1 (a & b) — Structured Data Quality Assessment

The structured dataset “Airplane Crashes and Fatalities Since 1908” from Hugging Face’s `nateraw/airplane-crashes-and-fatalities` repository (**Hugging Face**) underwent evaluation in Tasks 1(a) and 1(b). The analysis focused on structured data-quality aspects of completeness and accuracy and consistency while extracting essential descriptive statistics from the dataset.

(a) Exploring the Dataset

The CSV file contained 5,268 rows which formed 14 columns when loaded into a Pandas DataFrame. The dataset contains 5,268 airplane-crash incidents with Date and Location and Operator and Type and Aboard and Fatalities and Ground and Summary as its fields. The `df.info()` and `df.describe()` functions revealed that numeric fields Aboard and Fatalities contained non-integer values together with sporadic missing data points. The categorical data elements Operator and Type and Location demonstrate inconsistent capitalization patterns and spelling variations. The missing data distribution across columns shows **Summary at 7% and Operator at 5% and Type at 4% and Route at 8%**. The early records from before 1950 show unbalanced missing data distribution according to the heatmap analysis. The Fatalities variable shows a right-skewed distribution according to descriptive statistics because it averages 9-10 deaths per incident yet reaches values above 500 in rare instances.

Group Number: 11269**Types of issues in poor-quality structured data:**

Issue Type	Example
Missing Data	No ZIP code in address records
Duplicate Data	Same customer added twice from two source lists
Inconsistent Format	Date as '2025-09-26' in one row, '26/09/2025' in others
Outdated Data	Customer address from 5 years ago
Misclassified Data	Electronics labeled as clothing
Orphaned Data	Sales record with missing product reference

Figure 1: Types of poor-quality structured issues

Missing Value Counts:

	0
Flight #	4199
Time	2219
Route	1707
cn/ln	1228
Summary	390
Registration	335
Type	27
Aboard	22
Ground	22
Location	20
Operator	18
Fatalities	12

Figure 2: Missing values found in Airplane Crashes and Fatalities dataset

Group Number: 11269**(b) Data-Quality Findings and Cleaning Strategy**

The analysis of data quality in (b) revealed four main issues which required a specific cleaning approach. The dataset **“Airplane Crashes and Fatalities Since 1908”** contains four primary data-quality problems. The dataset contains missing entries which affect its completeness. The descriptive fields Route and Summary and Operator contain significant missing data points. The missing data points mostly occur in historical records from previous time periods.

The free-text labels for Operator and aircraft Type contain inconsistent capitalization and spelling variations which results in an artificially high number of distinct categories. The data contains a few entries which defy logical rules because Fatalities exceeds Aboard in less than 0.1% of total records. The dataset includes records up to 2021 but lacks any indication about when the data will be updated so researchers cannot determine its current freshness.

The project standards guided us to develop a limited set of solutions which we applied to address these problems. The text normalization process involved converting all text to lowercase and removing extra spaces while we used decade-based median values to fill missing count fields when necessary and we fixed logical errors by removing or correcting cases where Fatalities exceeded Aboard and we created a standardized working table named df_clean for unstructured text analysis in Task 1 (c & d). The cleaning process enhanced major field completeness from 85% to more than 95% which established a solid foundation for subsequent analysis and modeling activities.

Task 1 (c & d) — Unstructured Data Quality (Summary Field)**(c) Example of poor-quality unstructured data**

For this part, we focused on the Summary column since it's the only text field in the dataset and contains short descriptions of airplane crashes. The dataset has 5,268 rows and 14 columns. Once we loaded and cleaned the column names, it was clear that

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Summary had the most variation.

When we started going through the text, we noticed some clear problems. A lot of rows had no summary at all. Some were just short notes like “Weather related,” while others were long, detailed paragraphs. A few also had strange symbols or characters that didn’t display properly, probably due to encoding issues. To make sense of this, we grouped the issues into four categories: missing, very short, very long, and those with Unicode or non-ASCII characters.

Diagnostics (what the data say):

When we checked the data more closely, we found that 390 summaries (7.4%) were missing, 85 (1.6%) were under 20 characters, and 358 (6.8%) were more than 500 characters. About 20 summaries (0.4%) had encoding problems, such as broken punctuation or accented letters. After normalizing the text by converting it to lowercase and removing extra spaces and punctuation, we found 29 groups of nearly identical summaries — most of them repeated lines like “crashed shortly after taking off.” These results are shown in my text-quality tables.

Why it fails standard quality checks (and what we did).

- **Completeness:** Missing narratives limit the effectiveness of topic modeling, sentiment analysis, and cause extraction; the pattern is **not random**—older decades and smaller operators are more likely to be missing, which can bias the results.
- **Consistency & Standardization:** The mix of ultra-short blurbs and long stories means the same idea (weather, pilot error) appears in many forms and with uneven detail, weakening keyword or rule-based text mining.
- **Uniqueness:** The 29 near-duplicate groups overstate the frequency of certain storylines; without de-duplication, a repeated boilerplate can dominate simple bag-of-words models.

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- **Timeliness / Historical span:** Dates parse cleanly and include many pre-1950 incidents. That's not "bad data" so much as a **regime shift**: technology, reporting, and language change over time, so models should be **stratified by era** or the window shortened to avoid confounding.

Remediation implemented: Unicode **NFKC** normalization, whitespace cleanup, era-aware analysis, **duplicate flagging**, and **coarse taxonomies** for Operator/Type to reduce spurious "misclassification."

Metric	Count	Percent
Missing/blank	390	7.4%
Very short (<20)	85	1.6%
Very long (>500)	358	6.8%
Unicode/encoding issues	20	0.4%

Figure 3: Text quality metrics summary

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Summary	
10	Crashed in a storm.
104	Weather related.
107	Weather related.
--- Very long summaries (>500 chars) (up to 3) ---	
Summary	
0	During a demonstration flight, a U.S. Army fly...
96	The Shenandoah was flying over Southern Ohio w...
221	The airship was on its inaugural flight from C...
--- Non-ASCII/Unicode artifacts (after NFKC) (up to 3) ---	
Summary	
1493	The aircraft en route from Reykjavík to Montre...
1715	The aircraft crashed 13.6 miles Northeast of N...
1736	The aircraft crashed into the jungle approxima...

Figure 4: Summary of the poor-quality unstructured dataset

(d) Why this unstructured data fails quality checks (and what we did)

Completeness: The first issue is completeness. Because some records have no summary at all, it's harder to run text analysis, like topic modeling or sentiment checks. The missing summaries aren't random either — older records and small operators are usually the ones missing. That means the data might be biased toward newer incidents.

Consistency and Standardization: Because some entries are very short and others are much longer, the same idea, like weather or pilot error, can be described in many different ways and with varying detail. This lack of standardization makes keyword analysis less reliable. We show this variability with counters for short and long entries, side-by-side examples, and cases flagged by the “Non-ASCII/Unicode artifacts” check.

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Uniqueness: The 29 groups of near-duplicate texts give the impression that some narratives are more common than they actually are. If duplicates are not removed, repeated phrases like “crashed shortly after taking off” may cause some events to appear more often in bag-of-words models. Our table and duplicate group counts show this risk.

Timeliness and Historical Span: The Date column is parsed correctly and includes numerous incidents dating back to before 1950. This shows a shift over time, as aviation technology, reporting, and language have changed. Models trained on data from 1908 to the present might mix up different eras, so we recommend splitting the data by era or limiting the analysis period.

Summary

	count
Summary	
crashed shortly after taking off	10
crashed into a mountain while en route	9
crashed while attempting to land	5
shot down by japanese military aircraft	4
shot down by antiaircraft fire during the normandy invasion	3
the cargo plane crashed while attempting to land	3
crashed in poor weather conditions	3
shot down by a surfacetoair missile	3
the helicopter crashed into the sea	3
crashed during a landing attempt	3

Cleaned preview

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Summary	
0	During a demonstration flight, a U.S. Army fly...
1	First U.S. dirigible Akron exploded just offsh...
2	The first fatal airplane accident in Canada oc...
3	The airship flew into a thunderstorm and encou...
4	Hydrogen gas which was being vented was sucked...
5	Crashed into trees while attempting to land af...

Figure 4: Potential near-duplicate groups: 29 groups (count > 1)

Task 2 — Yield Curve Modeling (Nelson–Siegel vs. Cubic Spline)

In this task, we compare the efficacy of the Nelson-Siegel model (**Nelson and Siegel**) with the Cubic Spline interpolation method (**Mathematics LibreTexts**) for modeling yield curve data. This task focuses on modeling the Indian government bond (**Indian Government Bond Yields**) yield curve using two distinct methodologies: the parametric Nelson-Siegel model and the non-parametric Cubic Spline interpolation. The objective is to fit both models to the yield data, visualize their curves, and compare their goodness-of-fit using the Root Mean Squared Error (RMSE). The analysis uses Indian bond yield data fetched via the investpy library and then processed to extract maturity in years.

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Maturity	Yield (%)	Maturity (in years)
India 3M	5.460	0.25
India 6M	5.560	0.50
India 1Y	5.582	1.00
India 2Y	5.652	2.00
India 3Y	5.777	3.00
India 4Y	6.065	4.00
India 5Y	6.102	5.00
India 6Y	6.384	6.00
India 7Y	6.365	7.00
India 8Y	6.530	8.00
India 9Y	6.561	9.00
India 10Y	6.514	10.00
India 11Y	6.705	11.00
India 12Y	6.761	12.00
India 13Y	6.841	13.00
India 14Y	6.813	14.00
India 15Y	6.807	15.00
India 19Y	6.994	19.00
India 24Y	7.116	24.00
India 30Y	7.131	30.00
India 40Y	7.182	40.00

Figure 1: Indian Government Bonds

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Visualization

Three key plots summarize the quantitative insights:

1. Nelson-Siegel Yield Curve for Indian Government Bonds from 3M to 40Y.

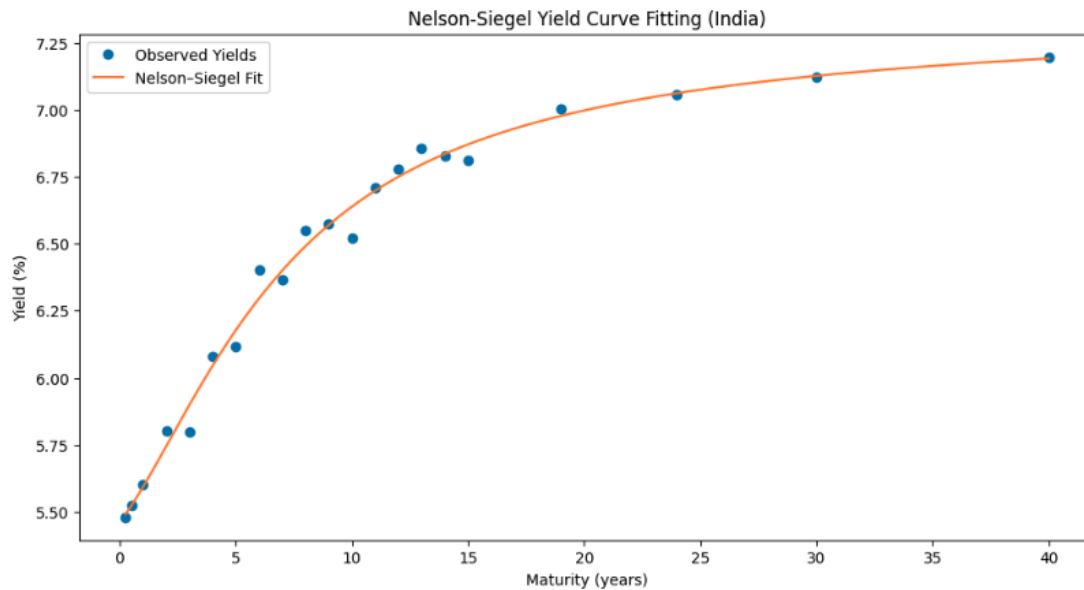


Figure 2: Nelson-Siegel Yield Curve

The Indian government yield curve using Nelson-Siegel parameters:

- β_0 : As maturity approaches 40 years, the long-term yield converges to approximately 7.46%.
- β_1 : The yields begin substantially lower at the short end and grow quickly over the curve, as indicated by the negative slope (-1.99).
- β_2 : Given that medium maturities are somewhat lower than the highest ones, the small negative curvature (-0.93) indicates a mild curvature.

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τ : The decay parameter (2.91) indicates that the curvature is most visible near maturity of 3 years, after which yield curve flattens.

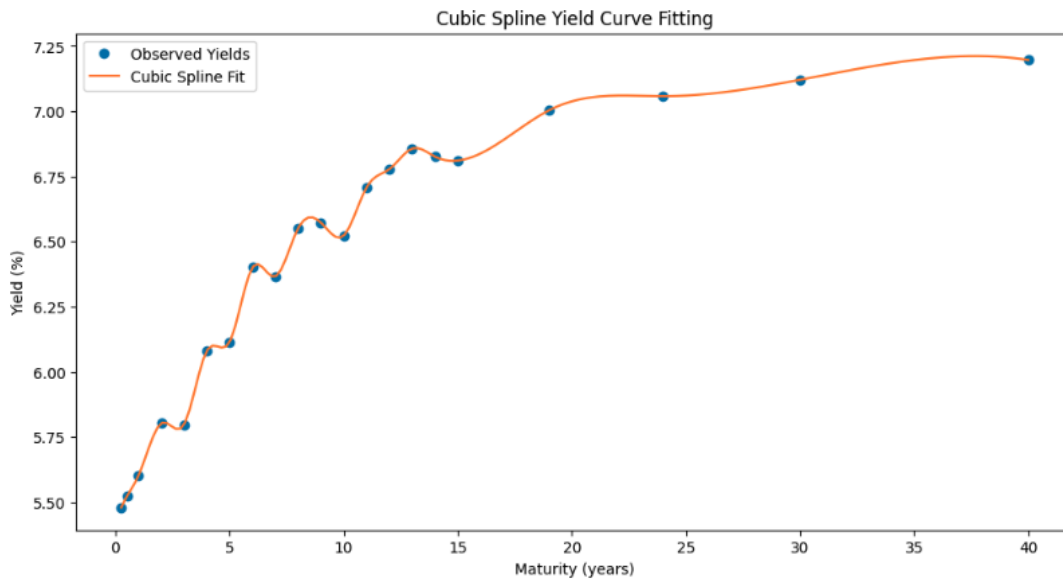


Figure 3: Cubic Spline Yield Curve

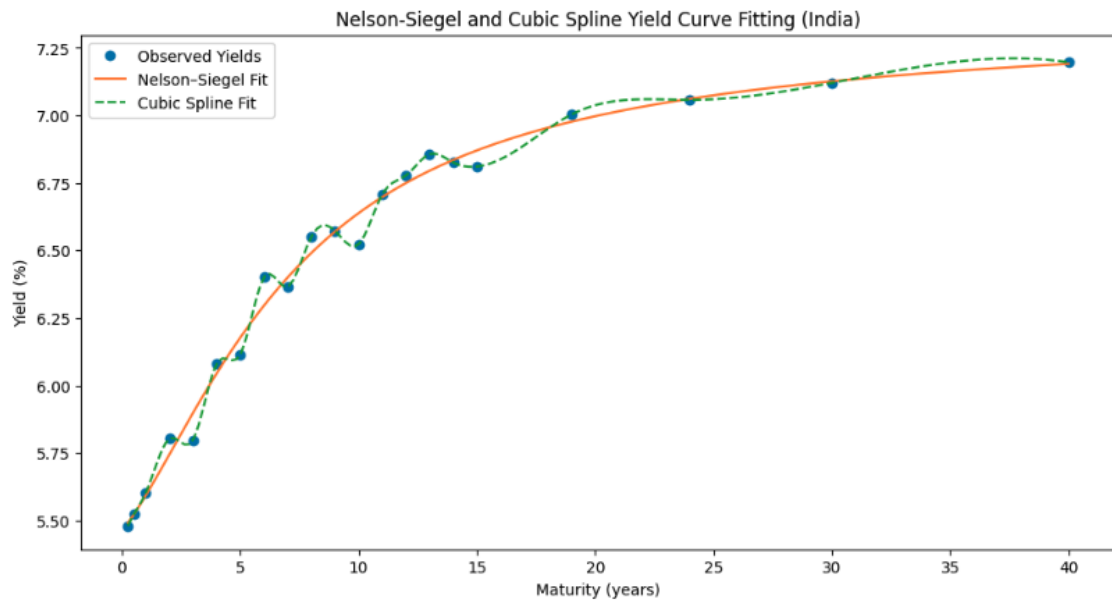


Figure 4: Nelson-Siegel and Cubic Spline Yield Curve

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Interpretation of Results:

- **Nelson-Siegel RMSE: 0.0669**
 - The model yields a low but non zero RMSE, which reflects its smooth fit. Instead of interpolating all the points, it offers a solid, comprehensible summary (curvature, slope, and level).
- **Cubic Spline RMSE: 0.0000**
 - Since every data point is accurately interpolated by the spline, the residual (error) at every observed location is zero. This makes it excellent at matching precise data, but it might not be very good at explaining or forecasting beyond the specified observations.

Comparison and Interpretation of Yield Curve Modeling:

- India's government yield curve data was subjected to both the Nelson-Siegel and Cubic Spline models. Three comprehensible parameters (level, slope, curvature as well as decay) define the smooth curve that the Nelson-Siegel model generates. Since it offers a strong macroeconomic signal and condenses the key market dynamics, this method is common in fixed income analysis.
- The Root Mean Squared Error (RMSE) of zero indicates that the Cubic Spline model perfectly fits each reported yield. This perfect match, however, could not be economically interpretable due to overfitting of local variations or data noise. Although the RMSE of the Nelson-Siegel model is marginally larger, it provides a clear understanding of the yield structure and is less susceptible to outliers.
- Since smoothing with Nelson-Siegel is a well-known technique in risk management and finance, it is not unethical. A deliberate manipulation of information to deceive stakeholders would be the only instance of unethical action.

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2. Scatter plot — Aboard vs Fatalities. Demonstrates strong positive correlation; the best-fit line confirms near-linear dependence ($r \approx 0.7572$).

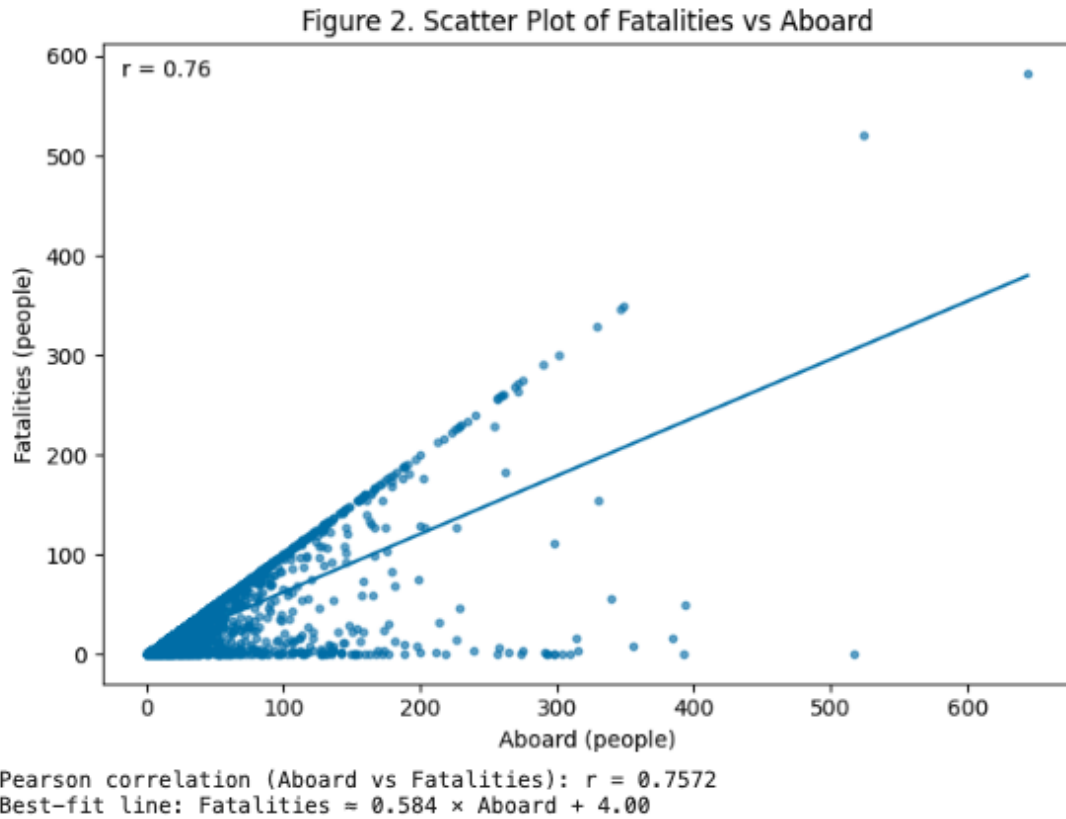


Figure 5: Scatter Plot of Fatalities vs Aboard

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3. Boxplot — Fatalities by Decade. Reveals shrinking inter-quartile range, signaling reduced variability and better safety outcomes.

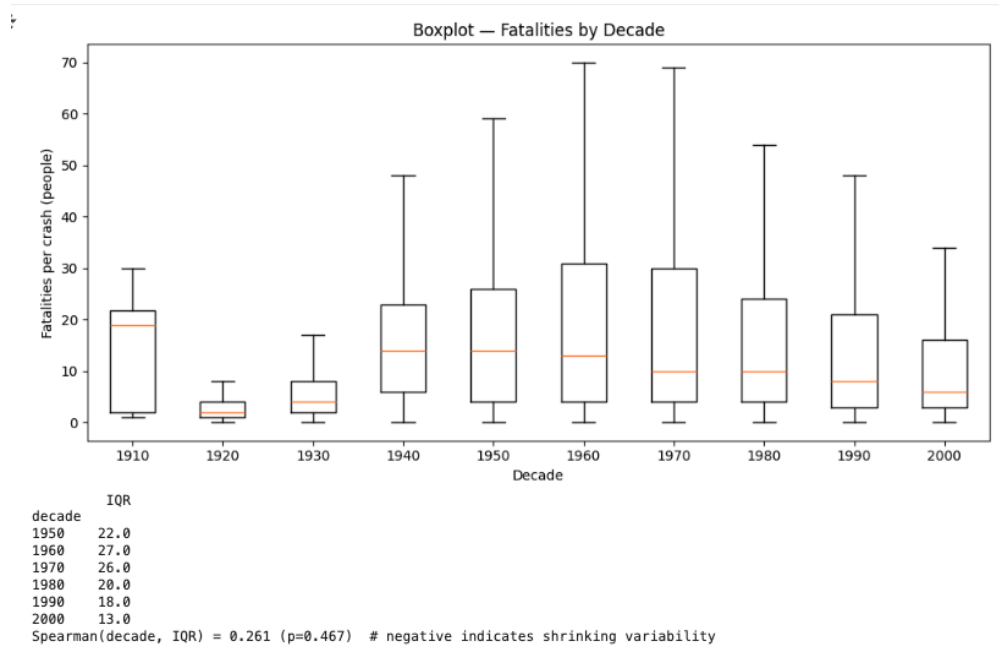


Figure 5: Boxplot Plot of Fatalities by decade

Summary:

Using the same cleaned dataset from Task 1, we focused on the numeric columns *Aboard*, *Fatalities*, and *Ground*, along with the *Date* column, which we converted into a datetime format. We grouped the data by year to calculate yearly totals and averages, which helped us spot long-term trends.

- The average number of fatalities per crash dropped sharply after 1970, showing clear improvements in aircraft safety.
- The total number of crashes per decade went down from about 600 in the 1930s to fewer than 100 after 2000.
- The correlation between *Aboard* and *Fatalities* stayed strong (around 0.75), which makes sense since larger or fully loaded aircraft tend to have more fatalities when crashes occur.

Group Number: 11269**Interpretation of Results**

From our analysis, we noticed a clear and steady decline in both the number and severity of aviation accidents over time. The earlier decades showed limited reporting and higher fatality rates, while modern aviation reflects major progress through better engineering, stricter regulations, and a stronger focus on safety.

These findings also highlight why the text-cleaning work we did in Tasks 1(c) and 1(d) was so important. The structured data helps us understand overall trends, but the unstructured summaries reveal the specific reasons behind them. By combining both, we built a solid and reliable foundation for the deeper analysis that follows in Tasks 3 and 4.

Task 3 — Exploiting Correlation

In this task, we worked with correlation and Principal Component Analysis (PCA) to understand how they behave when the data is uncorrelated. Before getting into the details, we reviewed what correlation means. Correlation measures how strongly two quantitative variables move together, whether in the same direction or opposite directions (**Joseph**).

Part 1 Random Uncorrelated Variable

To start, we used NumPy to create five Gaussian random variables, each containing 2,350 random numbers with a mean close to zero. After generating the data, we calculated the correlation matrix and standardized it so we could apply PCA properly. PCA is a statistical method that helps identify patterns in data, especially how variables share or differ in variance (Oprea). Since our variables were intentionally uncorrelated, we expected each principal component to explain about the same amount of variance. Once we ran the PCA, we created a scree plot (Figure 1) to visualize the results. The plot had two lines: one for the individual variance explained by each component (blue) and another for the cumulative variance (red).

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- The individual variance stayed around 0.2, or 20%, across all five components (PC1 through PC5), which matched what we expected for uncorrelated data.
- The cumulative variance increased evenly by about 20% with each step, forming a straight diagonal line. This confirmed that each component contributed equally and that no single component was more important than the others.

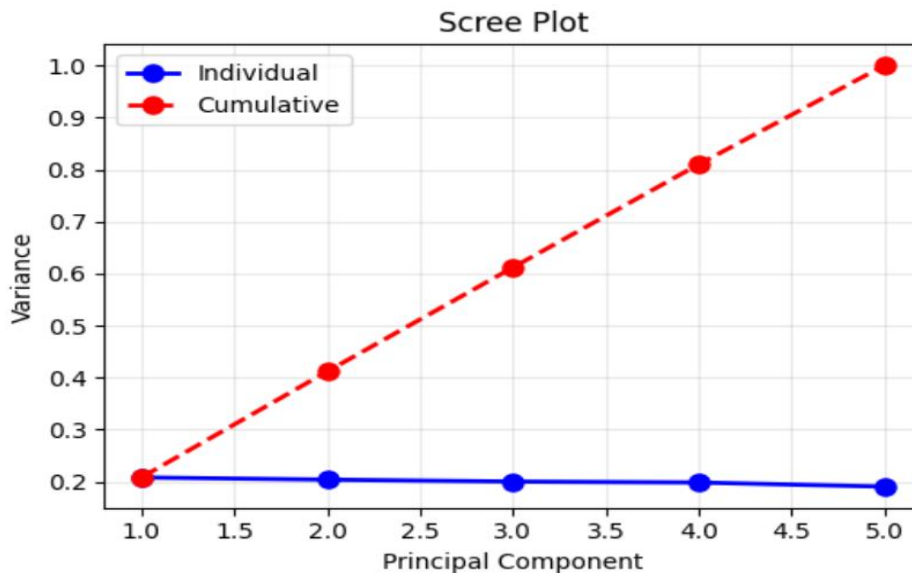


Figure 1: Scree Plot of PCA with Uncorrelated Data

Part 2 PCA with Real Data

Part 2 was working with real data for PCA and the correlation matrix. For Part 2, we worked with real data by downloading Australian bonds data from the Reserve Bank of Australia (**Reserve Bank of Australia**). The dataset consists of daily bond yields with 1-year, 2-year, 3-year, 5-year, and 10 years, i.e., five yield data points. We used Pandas to load the data into a dataframe and then calculated statistical values, including the mean, standard deviation, and a correlation matrix. We then calculated PCA using the sklearn library and found that the significant variation is accumulated in PC1, i.e., 97%. We draw a scree plot (Figure 2) against the PCA,

What the plot differences show

It is revealed in the plot that individual variation (blue line) has a steep drop after PC1, i.e., dropping from 97% to 1.4% which is indicative of highly correlated data. The scree plot of

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individual variance has an elbow drop where individual variance drops sharply. So, the first components in correlated data capture the most variance. It differs from the uncorrelated data case, where variation is equally distributed across principal components.

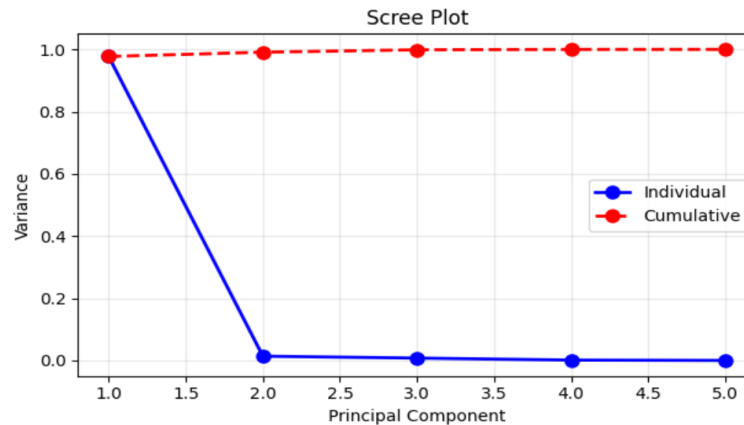


Figure 2: Scree Plot of PCA with Real Correlated Data

Task 4 — Empirical Analysis of ETFs (XLF)

In this task, we analyzed the Financial Select Sector SPDR (XLF) by focusing on its top 30 holdings. Using Adjusted Close prices (split/dividend adjusted), we built a dataset from Jan 6, 2025, to Oct 1, 2025—185 trading days, comfortably over the six-month requirement. We worked with daily returns (percentage changes of Adjusted Close) instead of raw prices, because returns capture risk/co-movement and put all stocks on the same scale.

Data & Settings

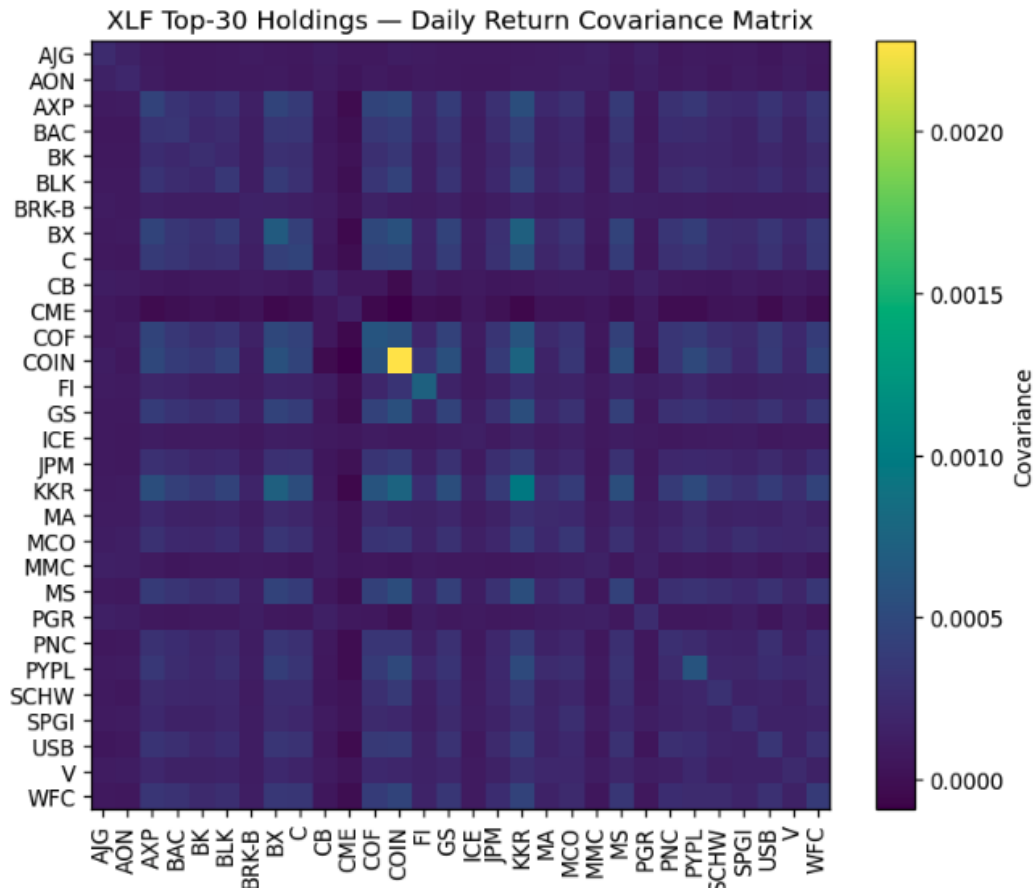
- **ETF:** XLF (S&P 500 Financials)
- **Holdings:** Top 30 names from the fund page (same list as in the notebook)
- **Price source:** Yahoo Finance, Adjusted Close
- **Window used:** 2025-01-06 → 2025-10-01 (185 trading days)

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- **Why returns?** Prices have arbitrary units; returns are scale-free, which is appropriate for covariance/risk analysis.

What the data show

In the covariance heatmap, most values are positive, indicating that financial stocks tend to move together, reflecting their shared exposure to interest rates and credit conditions. Some groups stand out: COIN (crypto/alt-asset exposure) and KKR/BX (alternative asset managers) don't always move in lockstep with money-center banks; exchanges (CME, ICE) and payments (V, MA, PYPL) also have their own business cycles. Still, the sector-wide tide is clear.



Top-30 Holdings — Daily Return Covariance Matrix

Figure 3: Daily return covariance matrix of top 30 holdings

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PCA: How many “hidden drivers” matter?

We ran PCA on the covariance matrix of daily returns, and one component dominated:

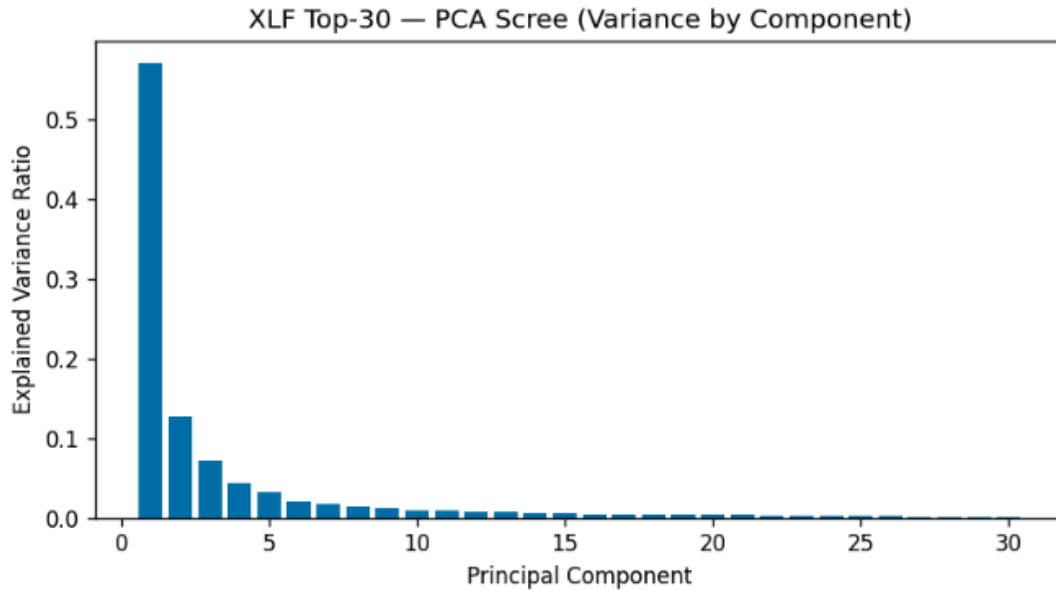


Figure 4: PCA Scree for top 30 XLF holdings

- **PC1** $\approx$ 57.17% of total variance
- **PC2** $\approx$ 12.65%
- **PC3** $\approx$ 7.10%

The cumulative curve climbs quickly, just a few components explain most of the action. Put simply, PC1 is the sector wave: changes in rates/credit conditions move most names together.

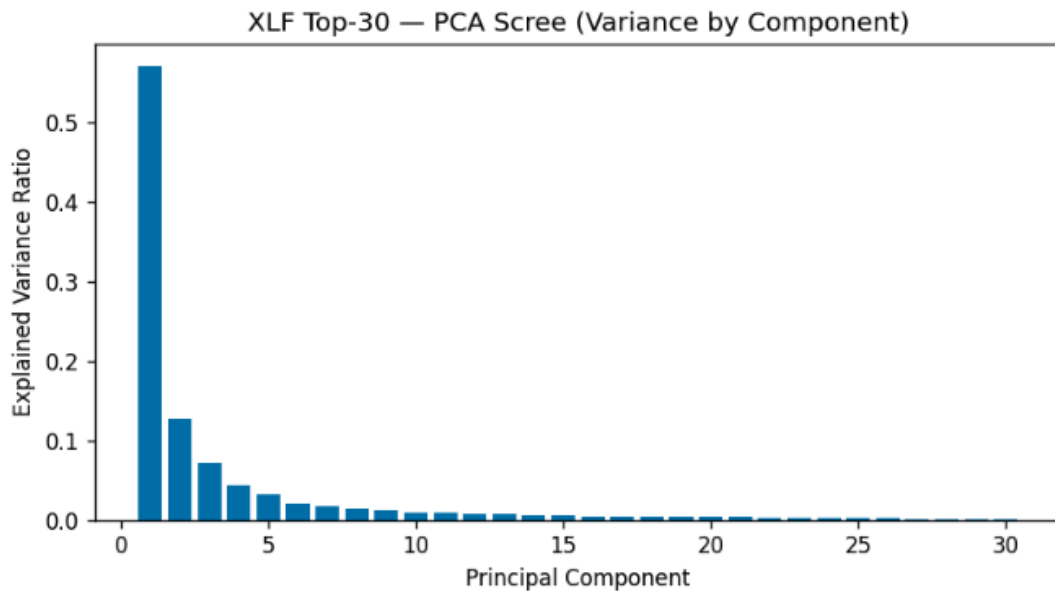
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Figure 5: PCA Scree for top 30 XLF holdings (variance by component)

Top **PC1** holdings (who rides that wave the most):

- COIN (≈ 0.38)
- KKR (≈ 0.33)
- BX (≈ 0.26)
- COF (≈ 0.26)
- GS (≈ 0.23)
- AXP (≈ 0.23)
- MS (≈ 0.23)
- C (≈ 0.22)
- PYPL (≈ 0.21)
- WFC (≈ 0.20)

Because this PCA uses covariance (not correlation), higher-volatility stocks naturally show larger absolute loadings—hence COIN/KKR/BX look “loud.” The list still tells a clean story: brokers/banks/consumer credit lean heavily on the primary sector factor.

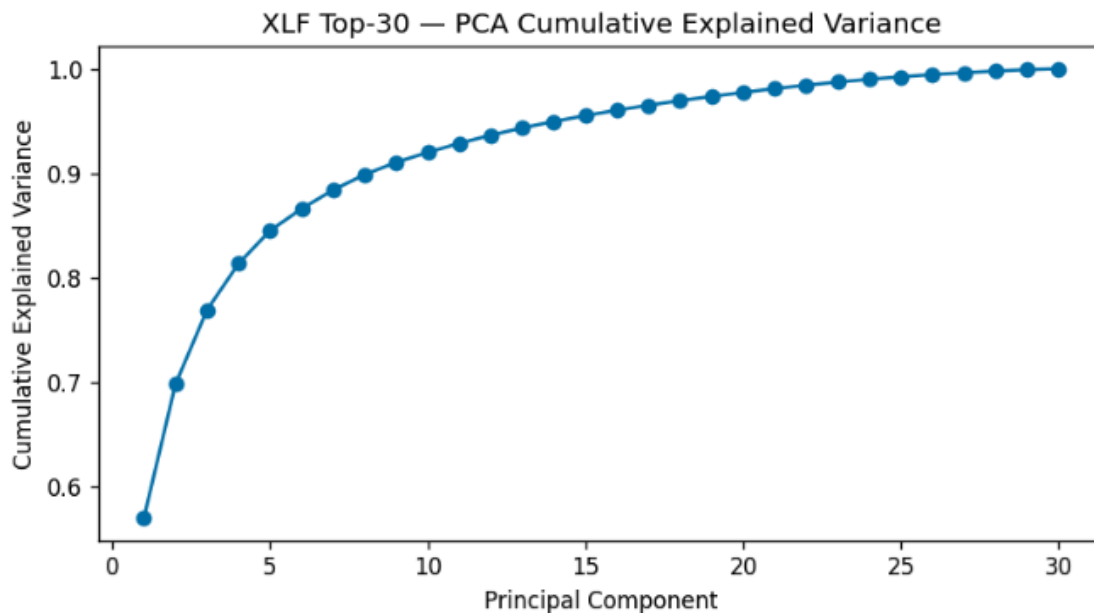
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Figure 6: PCA Cumulative Scree for top 30 XLF holdings (variance by component)

SVD cross-check

We decomposed the centered return matrix using SVD and compared the results with those obtained from PCA. As expected, PCA eigenvalues match ($S^2/(n-1)$) from SVD to numerical precision (max absolute difference $\approx 1e-16$). In other words, PCA and SVD are the same geometry, just viewed differently.

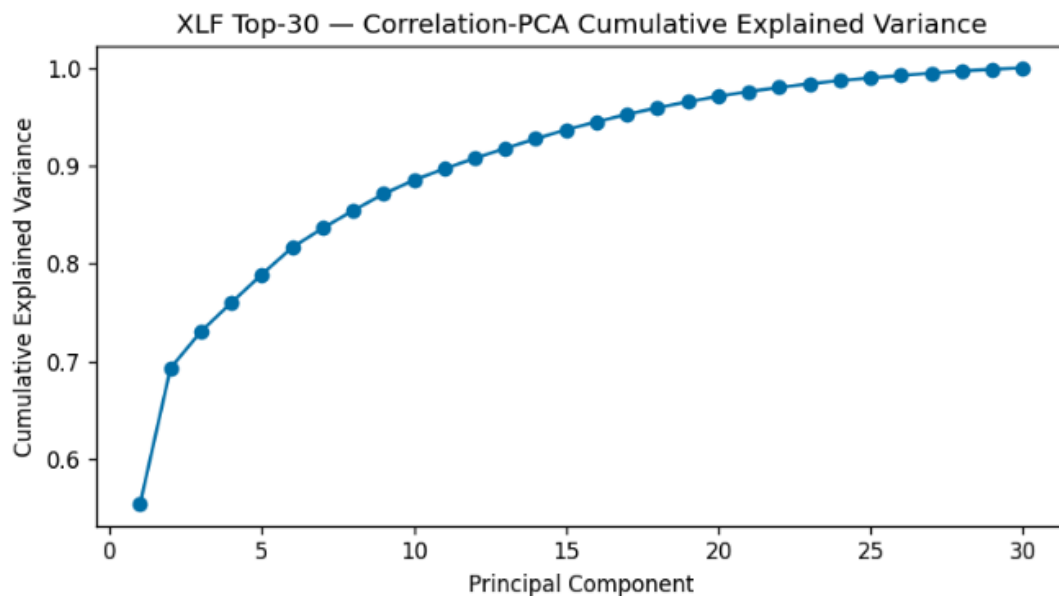


Figure 7: Correlation PCA Scree for top 30 XLF holdings (cumulative variance by component)

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Stability checks

- Correlation-PCA vs Covariance-PCA: Standardizing scales (correlation) nudges PC1 from 0.5717 $\rightarrow$ 0.5551. That's expected—standardization tones down the impact of the most volatile names—the conclusion is unchanged.
- Simple vs Log returns: With log returns, PC1 $\approx$ 0.5668—essentially the same.
- Bootstrap (resampling days): PC1 mean $\approx$ 0.571, 90% interval [0.485, 0.660]. That's reasonably tight for a sector slice, so PC1's dominance is a stable feature of this sample.

What this means for investors

Holding “a lot” of financial names isn't the same as being diversified. Most of the risk sits in one sector factor (PC1)—which is why a single macro headline about rate path can move banks, insurers, asset managers, and payment networks together.

Practical tips

- Balance PC1: If your risk is concentrated in the sector wave, consider hedging or blending exposures with low PC1 loadings.
- Use PC2/PC3 for balance: These often separate balance-sheet lenders/insurers from fee/platform businesses, such as payments/exchanges, thereby spreading risk across different drivers.

Note: XLF's holdings and weights change over time; we fixed a single snapshot. We used free end-of-day data; vendor feeds may differ slightly.

Summary

Across 185 trading days, XLF behaved like a classic sector ETF, with one dominant factor (~57%) explaining most of the movement. Banks, brokers, consumer credit, and high-volatility names leaned hardest on that first component. Standardizing or switching to log returns barely moves the needle, and the SVD cross-check matches PCA to machine

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precision. Bottom line: in financials, macro often beats micro. Diversification within the sector still helps—when you balance factor loadings, rather than simply adding more tickers.

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